

CLab 6

Stephanie Salgado

Setup

/ CLab6_Stephanie_Salgado /

Name	Last Modified
CreateSlice_CLab6.ipynb	5 minutes ago
Routing_with_IPv4_CLab6...	5 minutes ago

Since I set up my bastion key for the previous lab, I can go straight into JupyterHub. Once there, I create a folder where I will include all materials for this lab.

Reserving resources

```
[1]: # Import Fablib
from fabrictestbed_extensions.fablib.fablib import FablibManager as fablib_manager

fablib = fablib_manager()
fablib.show_config()

import json
import traceback
```

FABLIB Config									
Orchestrator	orchestrator.fabric-testbed.net								
Credential Manager	cm.fabric-testbed.net								
Core API	api.fabric-testbed.net								
Token File	/home/fablib/token.json								
Project ID	a70e2f5-9e12-4b9b-b412-0ea1a2c553b0								
Bastion Host	bastion.fabric-testbed.net								
Bastion Username	ssalga03_0000228084								
Bastion Private Key File	/home/fablib/work/fabric_config/bastion-key								
Slice Public Key File	/home/fablib/work/fabric_config/slice-key.pub								

Slice									
ID	1c0b0c4-98b3-4871-96a0-ad77f1c1b0e4								
Name	CLab6_Salgado_Stephanie_Salgado								
Lease Expiration (UTC)	2024-10-18 16:03:08 +0000								
Lease Start (UTC)	2024-10-17 16:03:08 +0000								
Project ID	a70e2f5-9e12-4b9b-b412-0ea1a2c553b0								
Data	testbed								

Nodes									
Name	Core	RAM	Disk	Image	Image Type	Host	Site	Username	Management IP
0-01	NDL_A	1	2	10	default/ubuntu_20	ssh-ndlib.fabric-testbed.net	EDUXX	ubuntu	2010-1ad-1700-206-1816-3a7f-1a75-7bae
0-02	NDL_A	1	2	10	default/ubuntu_20	ssh-ndlib.fabric-testbed.net	EDUXX	ubuntu	2010-1ad-1700-206-1816-3a7f-1a75-7bae
0-03	NDL_A	1	2	10	default/ubuntu_20	ssh-ndlib.fabric-testbed.net	EDUXX	ubuntu	2010-1ad-1700-206-1816-3a7f-1a75-7bae
0-04	NDL_A	1	2	10	default/ubuntu_20	ssh-ndlib.fabric-testbed.net	EDUXX	ubuntu	2010-1ad-1700-206-1816-3a7f-1a75-7bae
0-05	NDL_A	1	2	10	default/ubuntu_20	ssh-ndlib.fabric-testbed.net	EDUXX	ubuntu	2010-1ad-1700-206-1816-3a7f-1a75-7bae

Networks									
ID	Name	Layer	Type	Site	Subnet	Gateway	State	Error	
2470901-98b0-4450-98b3-45a0277209	lan_AB	L2	UBridge	EDUXX	None	None	Active		
181b0d4-2284-4186-9174-80e0c0f1a7	lan_BC	L2	UBridge	EDUXX	None	None	Active		
1810a3b-4056-4041-5541-2a728091015	lan_CA	L2	UBridge	EDUXX	None	None	Active		

Interfaces									
Name	Short Name	Mode	Network	Bandwidth	MTU	VLAN	MAC	Physical Device	IP Address
NDL_A-Gig-01	g1	NDL_A	lan_CA	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.1
NDL_A-Gig-02	g2	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.2
NDL_A-Gig-03	g3	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.3
NDL_A-Gig-04	g4	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.4
NDL_A-Gig-05	g5	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.5
NDL_A-Gig-06	g6	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.6
NDL_A-Gig-07	g7	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.7
NDL_A-Gig-08	g8	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.8
NDL_A-Gig-09	g9	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.9
NDL_A-Gig-10	g10	NDL_A	lan_AB	100	config	08:00:26:3A:C7:1E	enp7s0	enp7s0	10.10.10.10

Using the provided “CreateSlice_CLab6.ipynb” notebook, I start by importing Fablib. Then, I define and submit the slice.

Slice Name ^	Slice State	Lease End	Project	
IPv4_Routing_Stephanie_Salgado	StableOK	2024-10-18 11:03:08	Computer_Networks	 Slice ID

The slice can also be found under “My Slices”.

Setting up the network

```
#NDA
A_iface = NDA.get_interface(network_name='Lan_AB')
A_iface_ip_addr_add(addr='192.168.1.10', subnet=subnet1)
A_iface = NDA.get_interface(network_name='Lan_CA')
A_iface_ip_addr_add(addr='192.168.3.10', subnet=subnet3)

#NDB
B_iface = NDB.get_interface(network_name='Lan_AB')
B_iface_ip_addr_add(addr='192.168.1.11', subnet=subnet1)
B_iface = NDB.get_interface(network_name='Lan_BC')
B_iface_ip_addr_add(addr='192.168.2.11', subnet=subnet2)

#NDC
C_iface = NDC.get_interface(network_name='Lan_BC')
C_iface_ip_addr_add(addr='192.168.2.12', subnet=subnet2)
C_iface = NDC.get_interface(network_name='Lan_CA')
C_iface_ip_addr_add(addr='192.168.3.12', subnet=subnet3)

for node in slice.get_nodes():
    for interface in node.get_interfaces():
        stdout, stderr = node.execute(f'sudo ip link set dev { interface.get_device_name()} up')
        print(f'Network status at node {node.get_name()}')
        node.execute("ip a")
        node.execute("sudo sysctl -w net.ipv4.ip_forward=1")

except Exception as e:
    print(f'Fail: {e}')

Network status at node ND_A
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc fq_codel state UP group default qlen 1000
    link/ether fa:16:3e:75:7d:ec brd ff:ff:ff:ff:ff:ff
    inet 10.30.9.194/19 brd 10.30.31.255 scope global dynamic enp3s0
        valid_lft 85953sec preferred_lft 85953sec
    inet6 2610:108:1700:6:f016:3eff:fe75:7dec/64 scope global dynamic mngt6addr noprefixroute
        valid_lft 86364sec preferred_lft 14364sec
    inet6 fe80::f816:3eff:fe75:7dec/64 scope link
        valid_lft forever preferred_lft forever
3: enp7s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:b2:58:7a:c7:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.3.18/24 scope global enp7s0
```

I set up the network for the experiment by running the cell.

Installing “net-tools” and “traceroute” on each node

```
# Install Software
from ipaddress import ip_address, IPv4Address, IPv6Address, IPv4Network, IPv6Network

try:
    #remove existing routes for the experiment
    NDA.execute(f'sudo ip route flush dev {A_iface.get_device_name()}')
    NDC.execute(f'sudo ip route flush dev {C_iface.get_device_name()}')

    # install net-tools on each node
    for node in slice.get_nodes():
        #install route and traceroute
        node.execute("sudo apt update")
        node.execute("sudo apt install -y net-tools")
        node.execute("sudo apt install -y traceroute")

except Exception as e:
    print(f'Exception: {e}')

Get:10 http://security.ubuntu.com/ubuntu focal-security/multiverse Translation-en [3908 B]
Get:17 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [548 B]
Get:18 http://nova.clouds.archive.ubuntu.com/ubuntu focal/universe Translation-en [5124 kB]
Get:19 http://nova.clouds.archive.ubuntu.com/ubuntu focal/universe amd64 c-n-f Metadata [265 kB]
Get:20 http://nova.clouds.archive.ubuntu.com/ubuntu focal/multiverse amd64 Packages [144 kB]
Get:21 http://nova.clouds.archive.ubuntu.com/ubuntu focal/multiverse Translation-en [184 kB]
Get:22 http://nova.clouds.archive.ubuntu.com/ubuntu focal/multiverse amd64 c-n-f Metadata [9136 B]
Get:23 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3640 kB]
Get:24 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [358 kB]
Get:25 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [17.8 kB]
Get:26 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3330 kB]
Get:27 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [465 kB]
Get:28 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [548 B]
Get:29 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1237 kB]
Get:30 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [297 kB]
Get:31 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [28.3 kB]
Get:32 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [27.1 kB]
Get:33 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [7936 B]
Get:34 http://nova.clouds.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [1636 B]
```

Retrieving the slice

[illegible]

Once everything was configured, I could begin the experiment using the “Routing_with_IPv4_CLab6.ipynb” notebook. The first step was to import the “Fabric API” in order to retrieve the slice I created.

Accessing the nodes through ssh



Finally, I use ssh to log onto each node before starting the experiment.

Guided Experiment

Displaying current routing tables from each node

```
ubuntu@NDA:~$ route
Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0        0.0.0.0        255.255.224.0   U        0      0      0      0 enp3s0
169.254.169.254 10.30.6.11     255.255.255.255 UGH      100    0      0      0 enp3s0
192.168.1.0      0.0.0.0        255.255.255.0   U        0      0      0      0 enp8s0
ubuntu@NDA:~$

ubuntu@NDB:~$ route
Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0        0.0.0.0        255.255.224.0   U        0      0      0      0 enp3s0
169.254.169.254 10.30.6.11     255.255.255.255 UGH      100    0      0      0 enp3s0
192.168.1.0      0.0.0.0        255.255.255.0   U        0      0      0      0 enp7s0
192.168.2.0      0.0.0.0        255.255.255.0   U        0      0      0      0 enp8s0
ubuntu@NDB:~$ █

ubuntu@NDC:~$ route
Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0        0.0.0.0        255.255.224.0   U        0      0      0      0 enp3s0
169.254.169.254 10.30.6.11     255.255.255.255 UGH      100    0      0      0 enp3s0
192.168.2.0      0.0.0.0        255.255.255.0   U        0      0      0      0 enp7s0
ubuntu@NDC:~$ █
```

Adding a direct route from “NDA” to “NDC” (green route)

```
ubuntu@NDA:~$ ping 192.168.3.12
ping: connect: Network is unreachable
ubuntu@NDA:~$
```

A ping from “NDA” to the IP on “NDC” returns “Network is unreachable”.

```
ubuntu@NDA:~$ sudo route add 192.168.3.12 dev enp8s0
ubuntu@NDA:~$ route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0      0.0.0.0         255.255.224.0   U        0      0        0 enp3s0
169.254.169.254 10.30.6.11      255.255.255.255 UGH      100    0        0 enp3s0
192.168.1.0     0.0.0.0         255.255.255.0   U        0      0        0 enp7s0
192.168.3.12    0.0.0.0         255.255.255.255 UH       0      0        0 enp8s0
ubuntu@NDA:~$ ping 192.168.3.12
PING 192.168.3.12 (192.168.3.12) 56(84) bytes of data.
^C
--- 192.168.3.12 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2044ms

ubuntu@NDA:~$
```

Using the command “sudo route add 192.168.3.12 dev enp8s0”, should add the connection from “NDA” to “NDC” which can be confirmed by the new routing table. However, notice the ping fails once again. This is because the connection is only from “NDA” to “NDC” and not from “NDC” to “NDA”.

```
ubuntu@NDC:~$ sudo route add 192.168.3.10 dev enp7s0
ubuntu@NDC:~$ route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0      0.0.0.0         255.255.224.0   U        0      0        0 enp3s0
169.254.169.254 10.30.6.11      255.255.255.255 UGH      100    0        0 enp3s0
192.168.2.0     0.0.0.0         255.255.255.0   U        0      0        0 enp8s0
192.168.3.10    0.0.0.0         255.255.255.255 UH       0      0        0 enp7s0
ubuntu@NDC:~$
```

```
ubuntu@NDA:~$ ping 192.168.3.12 -c 1
PING 192.168.3.12 (192.168.3.12) 56(84) bytes of data.
64 bytes from 192.168.3.12: icmp_seq=1 ttl=64 time=0.082 ms

--- 192.168.3.12 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.082/0.082/0.082/0.000 ms
ubuntu@NDA:~$
```

Using the command “sudo route add 192.168.3.10 dev enp7s0” from “NDC” should fix it. This time, the ping is successful.

Assignment (red route)

Using “traceroute 192.168.2.12” before static routes

```
ubuntu@NDA:~$ traceroute 192.168.2.12
traceroute to 192.168.2.12 (192.168.2.12), 30 hops max, 60 byte packets
connect: Network is unreachable
ubuntu@NDA:~$
```

Since the static routes haven't been configured, the IP address “192.168.2.12” on “NDC” can't be reached from “NDA”.

Routing from “NDA” to “192.168.2.12” through “NDB”

```
ubuntu@NDA:~$ sudo route add -net 192.168.2.0 netmask 255.255.255.0 gw 192.168.1.11
ubuntu@NDA:~$ route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0      0.0.0.0         255.255.224.0   U      0      0        0 enp3s0
169.254.169.254 10.30.6.11      255.255.255.255 UGH    100    0        0 enp3s0
192.168.1.0    0.0.0.0         255.255.255.0   U      0      0        0 enp8s0
192.168.2.0    192.168.1.11    255.255.255.0   UG     0      0        0 enp8s0
192.168.3.12   0.0.0.0         255.255.255.255 UH     0      0        0 enp7s0
```

Using “sudo route add -net 192.168.2.0 netmask 255.255.255.0 gw 192.168.1.11” on “NDA” to send packets for the “192.168.2.0/24” network to “192.168.1.11”(NDB).

```
ubuntu@NDC:~$ sudo route add -net 192.168.1.0 netmask 255.255.255.0 gw 192.168.2.11
ubuntu@NDC:~$ route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
10.30.0.0      0.0.0.0         255.255.224.0   U      0      0        0 enp3s0
169.254.169.254 10.30.6.11      255.255.255.255 UGH    100    0        0 enp3s0
192.168.1.0    192.168.2.11    255.255.255.0   UG     0      0        0 enp7s0
192.168.2.0    0.0.0.0         255.255.255.0   U      0      0        0 enp7s0
192.168.3.10   0.0.0.0         255.255.255.255 UH     0      0        0 enp7s0
ubuntu@NDC:~$
```

I used “sudo route add -net 192.168.1.0 netmask 255.255.255.0 gw 192.168.2.11” on “NDC” in order to direct packets for the “192.168.1.0/24” network to “192.168.2.11” (NDB).

```
ubuntu@NDB:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
```

Lastly, I used “sudo sysctl -w net.ipv4.ip_forward=1” on “NDB” to enable IP forwarding.

```
ubuntu@NDA:~$ ping 192.168.2.12
PING 192.168.2.12 (192.168.2.12) 56(84) bytes of data.
64 bytes from 192.168.2.12: icmp_seq=1 ttl=63 time=0.162 ms
64 bytes from 192.168.2.12: icmp_seq=2 ttl=63 time=0.125 ms
64 bytes from 192.168.2.12: icmp_seq=3 ttl=63 time=0.129 ms
64 bytes from 192.168.2.12: icmp_seq=4 ttl=63 time=0.132 ms
64 bytes from 192.168.2.12: icmp_seq=5 ttl=63 time=0.138 ms
^C
--- 192.168.2.12 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4083ms
rtt min/avg/max/mdev = 0.125/0.137/0.162/0.013 ms
ubuntu@NDA:~$
```

This time “ping 192.168.2.12” from “NDA” was successful.

Using “traceroute 192.168.2.12” after static routes

```
ubuntu@NDA:~$ traceroute 192.168.2.12
traceroute to 192.168.2.12 (192.168.2.12), 30 hops max, 60 byte packets
 1 192.168.1.11 (192.168.1.11)  0.207 ms  0.194 ms  0.190 ms
 2 192.168.2.12 (192.168.2.12)  0.296 ms  0.291 ms  0.273 ms
ubuntu@NDA:~$ █
```

This time the traceroute shows the route we added above. The first hop being from “NDA” to “NDB” and the second being from “NDB” to “NDC”.

Deleting the slice

IPv4_Routing_Stephanie_Salgado Dead

I run the final cell to delete the slice.

Summary

For this lab, I used FABRIC for Routing with IPv4. There were 3 nodes, "NDA", "NDB" and "NDC". The goal was to get "NDA" to ping two IPs on "NDC" using different routes. The "green route" did not work for me at first. However, after noticing there was an error in the command provided, I was able to get the ping to work. I also found success when using the "red route" configuration. I noticed the provided image had "ip_forward on" in "NodeB" which indicated I should make sure IP forwarding was enabled in "NDB". I enjoyed this lab because it felt very similar to other labs we have done using GNS3. I feel a lot more comfortable using the Linux commands now and was glad I was able to get the second ping to work.